

GLOBAL E-COMMERCE SALES & PROFIT INTELLIGENCE

Data Analytics & AI Internship Project

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Project report | Exploratory analysis, performance measurement, and business
recommendations

1. Executive Summary

This project analyzes a global e-commerce sales dataset to summarize commercial performance and identify areas for management review. The analysis covers sales, profit, order and customer activity, product categories, monthly patterns, customer segments, and regional results. The aim is to turn transaction-level data into concise indicators and practical follow-up questions—not to treat sales volume alone as the measure of business success.

The cleaned dataset contains **2,000** orders and **1,534** unique customers, generating **\$484,559.34** in sales and **\$158,872.32** in profit. The overall profit margin is **32.79%**. The output summary also reports **272** loss-making orders, making order-level profitability an important area for further investigation.

Key observations

- Furniture contributes the largest category profit in the output summary, while Office Supplies has the lowest category profit margin.
- Europe records the largest regional profit total; Middle East & Africa has the highest shipping-cost burden as a share of sales.
- Monthly sales vary substantially across the observed period, so month-to-month comparisons should be read as descriptive patterns rather than forecasts.
- The customer segment with the highest total profit is Consumer. These findings point to opportunities for margin review, shipping-cost investigation, and customer-level profitability analysis.

The report proceeds from dataset and method notes to KPI results, visual analysis, recommendations, limitations, and references.

2. Project Overview

2.1 Problem statement

E-commerce businesses need to understand how sales and profit are distributed across products, regions, customer segments, and time. A high sales total can conceal weak margins, high delivery costs, or loss-making orders. This project uses descriptive analytics to identify those patterns and propose questions and actions for business review.

2.2 Objectives

- Summarize overall sales, profit, margin, order volume, customer count, units sold, average order value, and average discount.
- Compare category and regional performance using sales and profitability measures.
- Review monthly sales patterns and identify high- and low-profit months in the available data.
- Translate the results into practical recommendations and clearly state analytical limitations.

2.3 Intended users

The intended audience includes e-commerce business managers, sales and operations teams, category managers, and analysts who need a concise performance overview to guide follow-up investigation.

3. Dataset and Methodology

3.1 Data source and scope

The project uses the Global E-Commerce Sales & Customer Data dataset selected for this analysis. The cleaned file supplied with the project contains **2,000** rows and **22** columns. The original row count was 2,000; the project summary reports that no duplicate rows were removed and 2,000 rows remained after cleaning.

Dataset source: [Kaggle — Global E-Commerce Sales & Customer Data](#). The source page should be consulted for the dataset's original description and usage terms.

3.2 Tools used

Python was used for data preparation, aggregation, and visualization. The project dependencies include pandas, NumPy, and Matplotlib. The generated outputs include cleaned transaction data, summary CSV files, chart images, and an HTML dashboard.

3.3 Analysis workflow

1. Load the transaction dataset and inspect its structure.
2. Prepare the data for analysis and export the cleaned dataset.
3. Calculate overall KPIs and grouped summaries for category, region, month, product, and customer segment.
4. Create charts to make comparisons and trends easier to interpret.
5. Review the resulting measures and translate them into management questions and recommended actions.

3.4 Data quality

The project's final insight output reports 2,000 original rows, zero duplicate rows removed, and 2,000 rows after cleaning. This confirms that the duplicate-removal step did not reduce the row count. It does not, by itself, establish that every field is complete, valid, or free of outliers; those checks should be documented separately if performed.

4. Key Performance Indicators

Indicator	Result	Interpretation
Total sales	\$484,559.34	Total recorded sales value in the analyzed dataset.
Total profit	\$158,872.32	Total recorded profit across analyzed transactions.
Overall profit margin	32.79%	Profit as a proportion of sales, as reported by the project.
Orders	2,000	Number of orders included in the summary.
Unique customers	1,534	Distinct customers represented in the dataset.
Units sold	7,115	Total quantity recorded.
Average order value	\$242.28	Average sales value per order, as reported.
Average discount	8.57%	Average discount reported in the summary.
Loss-making orders	272	Orders with negative profit; merits order-level review.

Interpretation: the headline totals provide a baseline for performance monitoring. The coexistence of positive overall profit and loss-making orders means aggregate profitability should be supplemented with an order-level view to locate the sources and causes of losses.

5. Product Category Analysis

The category chart compares sales totals across the four product categories. Furniture has the highest sales total, followed by Technology, Clothing & Accessories, and Office Supplies. The accompanying summary table adds profit and margin so that sales scale can be considered alongside profitability.

Category	Orders	Sales (USD)	Profit (USD)	Profit margin (%)
Furniture	507	256,274.68	81,171.57	31.67
Technology	567	139,518.22	48,268.65	34.60
Clothing & Accessories	413	69,375.63	26,112.94	37.64
Office Supplies	513	19,390.81	3,319.16	17.12

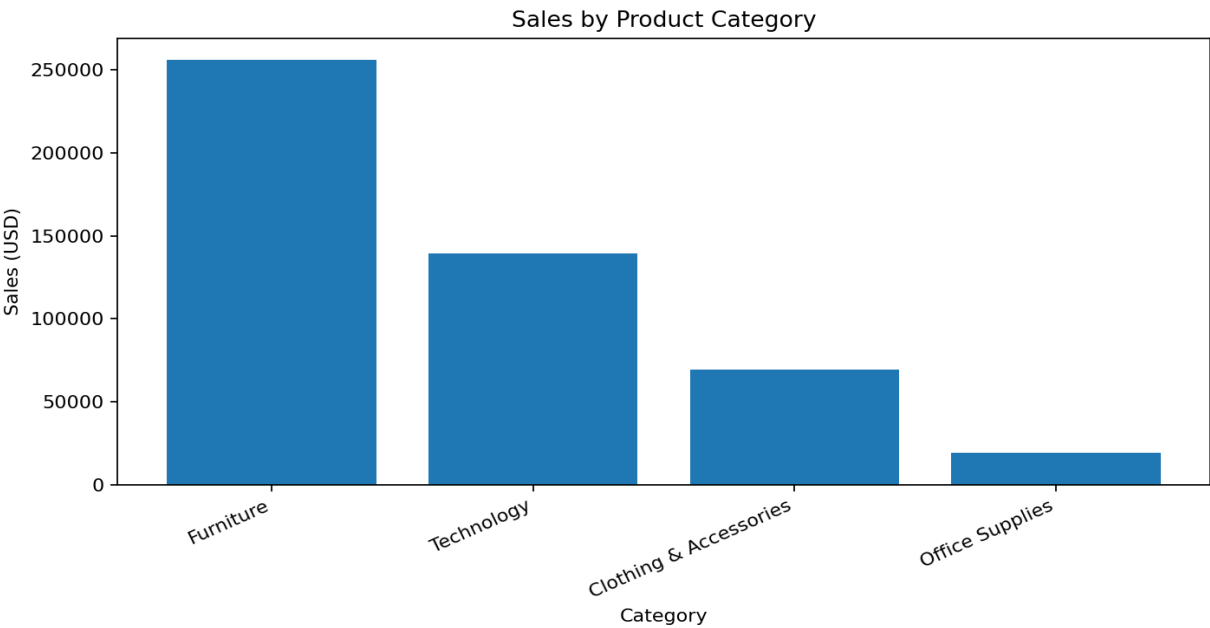


Figure 1. Sales by product category (USD). Bar heights show the category sales totals; the chart is descriptive and does not indicate margin.

Business interpretation: Furniture is the largest category by sales and by total profit in the summary, but its margin is below Clothing & Accessories. Office Supplies has the smallest sales total and the lowest reported category margin (17.12%). Review pricing, discounting, product mix, and cost structure in Office Supplies before pursuing volume growth alone.

6. Monthly Performance

The monthly sales trend shows substantial variation across months. Several months rise sharply relative to adjacent periods, and there are also pronounced dips. The plot is useful for spotting periods to investigate, but it does not explain the causes of variation; promotions, seasonality, inventory, product mix, and data coverage would need to be checked before attributing a cause.

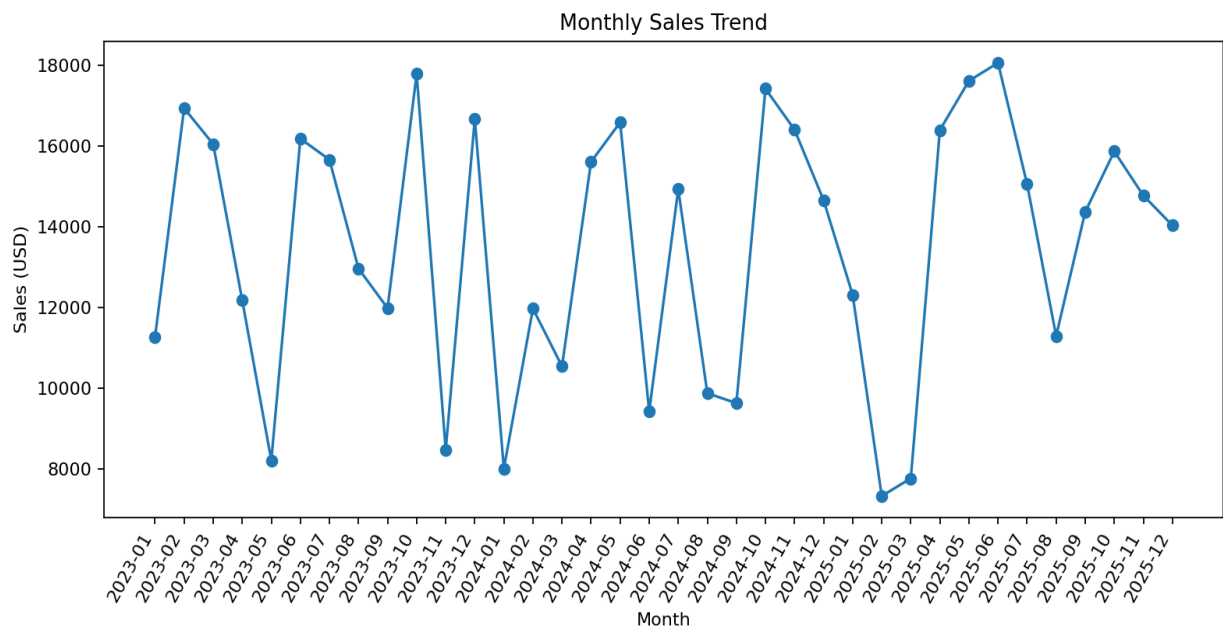


Figure 2. Monthly sales trend (USD). Each point represents the sales total for one month in the dataset.

The monthly summary identifies **2023-02** as the highest-profit month (5,980.06 USD) and **2025-02** as the lowest-profit month (2,430.75 USD). These are observed extremes in this dataset, not predictions for future periods.

7. Regional Performance

Regional comparison helps distinguish where profit is generated from where delivery costs may be taking a larger share of sales. Europe has the highest total profit in the summary, while Middle East & Africa has the highest reported shipping-cost burden.

Region	Orders	Sales (USD)	Profit (USD)	Margin (%)	Shipping / sales (%)
Europe	503	137,006.20	45,672.16	33.34	4.52
North America	578	133,876.38	45,250.09	33.80	4.34
Asia Pacific	520	121,707.51	39,116.61	32.14	6.19
South America	191	46,051.13	14,680.98	31.88	6.41
Middle East & Africa	208	45,918.12	14,152.48	30.82	7.25

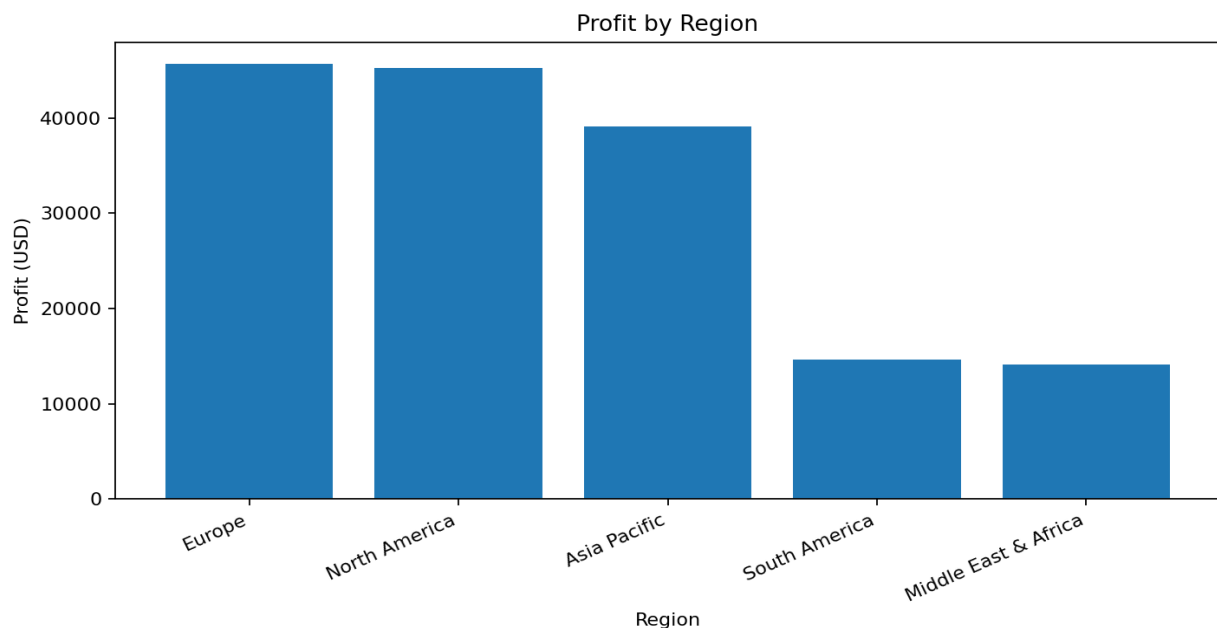


Figure 3. Profit by region (USD). The chart compares total profit, not profit margin or per-order performance.

Business interpretation: Europe and North America have similar total profit in the summary, while the Middle East & Africa region has a smaller profit total and the largest shipping-cost share of sales (7.25%). Investigate shipping routes, fulfillment methods, carrier rates, and delivery service levels in that region. Compare cost-to-serve and customer outcomes before changing service or pricing.

8. Customer Segment and Product-Level Review

The project summary identifies Consumer as the segment with the highest total profit (\$87,300.36). The segment-level table below provides the supporting comparison available in the generated output.

Customer segment	Orders	Sales (USD)	Profit (USD)	Margin (%)
Consumer	1006	256,287.74	87,300.36	34.06
Corporate	623	146,050.39	44,463.44	30.44
Home Office	371	82,221.21	27,108.52	32.97

A product-level summary was also generated. It can be used to examine whether high sales or profit is concentrated in a small number of products and to flag products that may warrant pricing, stock, or promotion review.

Product-specific conclusions should be based on the complete product summary rather than inferred from the category chart alone.

9. Recommendations

- 1. Review low-margin category performance.** Start with Office Supplies (17.12% reported margin). Break the result down by product, discount level, and cost components to identify whether the issue is concentrated or category-wide.
- 2. Investigate loss-making orders.** Review the 272 orders reported as loss-making. Check discount, shipping charges, returns or adjustments if available, order size, and product mix. Establish a repeatable exception report for negative-profit transactions.
- 3. Examine regional shipping burden.** Analyze the Middle East & Africa region's shipping cost as a share of sales (7.25%). Compare carriers, distance, fulfillment choices, order value, and delivery outcomes before selecting operational changes.
- 4. Monitor monthly movement with context.** Track sales and profit together, and annotate business events such as promotions or supply interruptions where reliable records exist. Avoid treating one unusually strong or weak month as a stable trend.
- 5. Use customer profitability and frequency for retention.** Segment customers by profit contribution, repeat purchase behavior, and average order value. Use these measures to inform retention or upsell experiments, then evaluate the results against margin as well as revenue.
- 6. Protect profitable activity while validating sustainability.** Maintain visibility into categories and regions contributing profit, but verify that their results remain consistent across time and are not driven by one-off transactions or unusually high discounts.

10. Limitations and Further Work

- The report is descriptive. It summarizes patterns in the supplied dataset and does not establish causal relationships or forecast future sales.
- The output summary records row counts and duplicate-removal results, but a complete field-by-field validation log (missing values, invalid dates, outliers, and correction rules) is not included in the supplied summary text.
- The dataset's represented period and geographic coverage should be interpreted according to its source documentation; results should not automatically be generalized to every e-commerce business or current market.
- The report does not claim that the recommendations have been implemented or that their financial impact has been measured.

Possible next steps include validating source definitions and currency conventions, adding a documented missing-value/outlier audit, analyzing discount and shipping effects at order level, tracking repeat-purchase cohorts, and testing any operational changes with a defined before-and-after measurement plan.

11. Conclusion

The analysis provides a compact view of e-commerce performance across overall KPIs, categories, months, regions, and customer segments. The results show that sales scale and profitability are not interchangeable: Office Supplies has the lowest reported category margin, loss-making orders are present, and shipping-cost burden differs by region. These findings can guide targeted investigation of pricing, discounts, delivery costs, and customer value. The next stage is to validate the underlying drivers at transaction level and track whether any changes improve profit without compromising customer outcomes.

12. References and Project Outputs

1. Kaggle. [Global E-Commerce Sales & Customer Data](#) (dataset source; access and usage terms should be checked on the source page).
2. Project-generated files: cleaned_ecommerce_sales.csv; kpi_summary.csv; category_summary.csv; monthly_summary.csv; region_summary.csv; segment_summary.csv; product_summary.csv; final_insights.txt; category_sales.png; monthly_sales_trend.png; region_profit.png; dashboard.html.

Appendix: KPI definitions

- Total sales: sum of the sales-value field across included records.
- Total profit: sum of the profit field across included records.
- Profit margin: total profit divided by total sales, expressed as a percentage.
- Average order value: total sales divided by the number of orders, as represented in the project summary.
- Shipping-cost burden: shipping cost expressed as a percentage of sales, as represented in the regional summary.